



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 2: Introduction

Patricia Galilea A.

ICS2523 Microeconomics

Lecture 2

Last lecture we

- Presented the course
- Started using Canvas
- Reviewed the **Opportunity Cost** concept

In today's lecture, we will start the introduction of the course:

- Give the first assignment
- Discuss the difference between Microeconomics and Macroeconomics
- Define Economic Models
- Compare Partial Equilibrium Models vs. General Equilibrium Models

Questions from last lecture

- Review the exercise about the opportunity cost.

Exercise

Suppose you won a free ticket to see Kid Voodoo concert tonight. You can't resell it.

Los Bunkers are performing on the same night, and their concert is the only other activity you are considering.

A Los Bunkers ticket costs \$40; on any given day, you would be willing to pay as much as \$50 to see them perform. In other words, if Los Bunkers tickets sold for more than \$50, you would pass on the opportunity to see him even if you had nothing else to do. There is no other cost of seeing either band.

What is your opportunity cost of attending Kid Voodoo concert?

Opportunity cost

What is the only thing **you would have to sacrifice** to attend Kid Voodoo concert?

You sacrifice seeing Los Bunkers concert

By not attending the Los Bunkers concert, you miss out a performance that would have been worth \$50 to you, BUT you also avoid having to spend \$40 for the ticket

So, the value of what you give up by not seeing Los Bunkers is:

$$\text{\$50} - \text{\$40} = \text{\$10}$$

If seeing Kid Voodoo worth at least \$10 to you, you should attend their concert. Otherwise, you should see Los Bunkers.

First Assignment

Read the second chapter of “The Economist Naturalist” (uploaded to our website). Inspired by their Q&A, create your own microeconomic question based on a "real-life" example and answer it using your microeconomic knowledge.

You should **upload your Q&A** in the **Written Assignment** Module from Canvas

Deadline: Monday, March 24th at 11.59 PM

Class assistantships: Survey

You prefer the class assistantships to be...

- a) Face to face (on Tuesdays at 14.50)
- b) Video Capsules
- c) Indifferent



Wooclap:

<https://app.wooclap.com/XTZMJF?from=event-page>

Introduction

Presentation

1 Microeconomics vs Macroeconomics

- **Microeconomics** looks at the smaller picture and focuses more on fundamental theories of supply and demand and how individual businesses decide how much of something to produce and how much to charge for it → focusing on **quantity (Q) and prices (P)**
- **Macroeconomics** looks at the big picture and focuses on the national economy. Provides a basic knowledge of how things work in the business world → focuses on **aggregate variables**

Understanding economics at both the micro and macro levels is essential for an overall perspective of how the economy works.

Introduction: Summary

Microeconomics	Macroeconomics
Equilibrium occurs when the quantity demanded equals the quantity supplied.	Equilibrium in an economy occurs when the aggregate demand equals the aggregate supply
There is a price for each good or service that will clear the market	There is a price level in an economy at which the aggregate demand will equal aggregate supply

2 Economic Models

Models are useful to **describe economic activities**

Economic models (and any kind of model) are abstractions from reality

Economic models provide aid in understanding economic behaviour

Economic models have **three features**:

1. *Ceteris paribus* assumption
2. Optimization assumption
3. Distinction: positive and normative analysis

2 Economic Models

Verification of economic models:

- Testing the validity of a model is a difficult task.
 - Are the model's assumptions reasonable?
 - Does the model explain real-world events?
- Methods used to verify economic models:
 - **The direct approach** → establishes the validity of the model's assumptions
 - **The indirect approach** → shows that the model correctly predicts real-world events

Activity 1

Profit-maximization model

Assumption:

- Firm seeks to maximize profits → simplification of reality

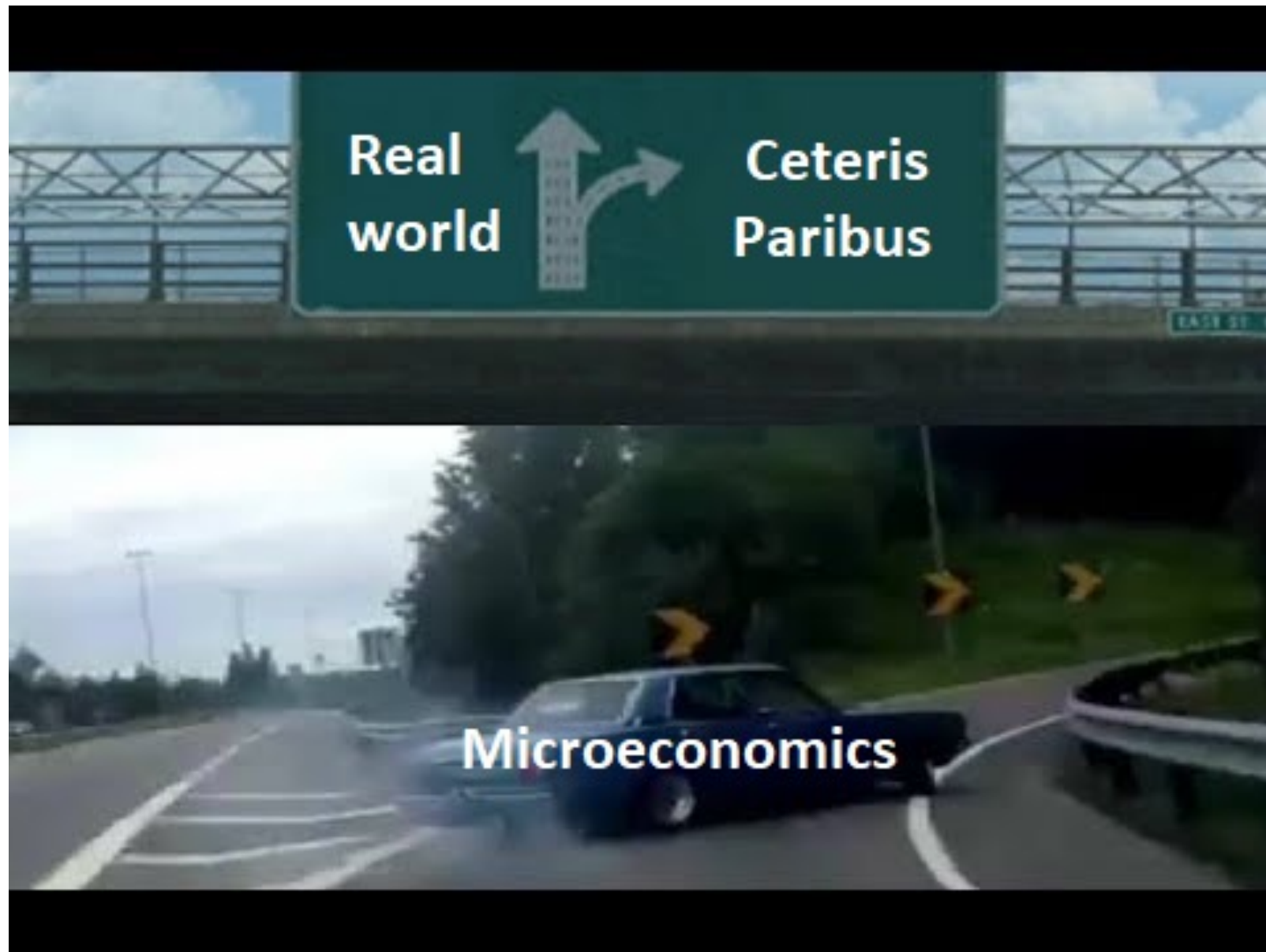
Is this simple model a good one?

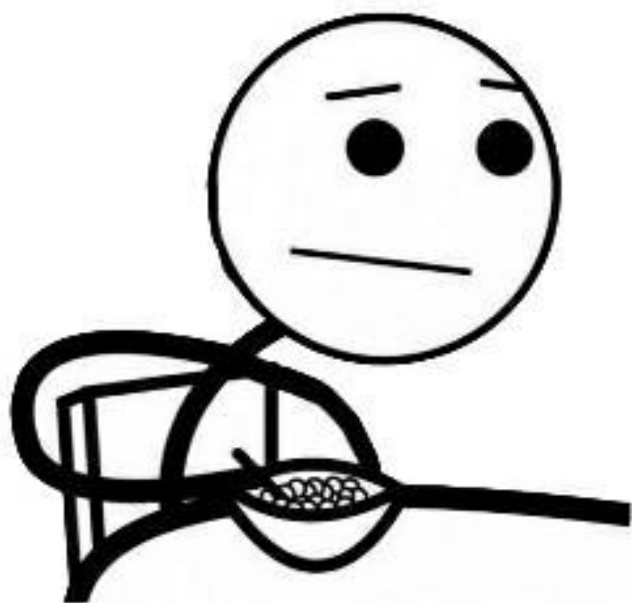
Discuss and test its assumptions (direct approach) and predictions (indirect approach)

2 Economic Models: General Features

1. *Ceteris Paribus*

- In any economic analysis, we usually assume that everything outside of the problem at hand remains constant
- When considering economics, it is helpful to first evaluate only two variables and then examine the effects of others upon the model
- Economists call this *ceteris paribus* or the **other things being equal assumption**





yeah its indifferent
to me, ceteris
paribus, wich
section to choose

Publicado el 10 mar. 2021



¡Bienvenid@s a Microeconomía en
inglés!

Publicado el 14 mar. 2021



I SAID CETERIS
PARIBUS!!

Activity 2

When discussing the laws of supply and demand, one could say that if the demand for a given product outweighs supply, *ceteris paribus*, prices will...

1. Rise
2. Fall
3. Remain constant



Wooclap:

<https://app.wooclap.com/XTZMJF?from=event-page>

Activity 2

An increase in the price of beef will result, ceteris paribus, in _____ beef being sold to consumers

1. Less
2. More



Wooclap:

<https://app.wooclap.com/XTZMJF?from=event-page>

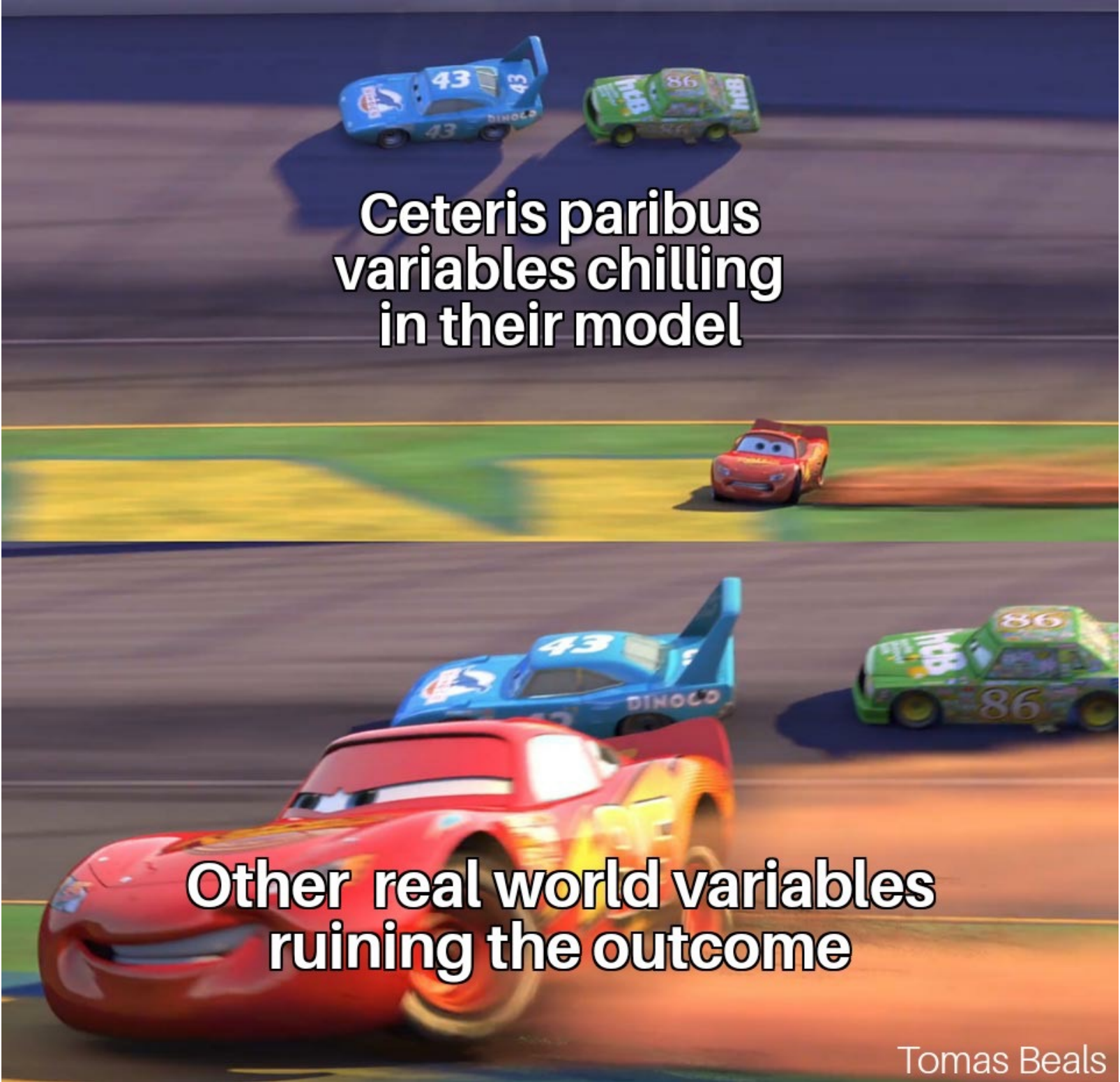
Ceteris Paribus final example

An increase in interest rates will *ceteris paribus* cause demand for loans to fall. Higher interest rates increase the cost of borrowing so there will be less demand for loans

However, what about if I do not assume *ceteris paribus*? Can we ensure that the demand for loans is going to fall?

If confidence is high (increasing), people might still want to borrow more

Ceteris paribus assumes things like confidence remain the same



**Ceteris paribus
variables chilling
in their model**

**Other real world variables
ruining the outcome**

2 Economic Models: General Features

2. Optimization Assumptions

Many economic models begin with the assumption that economic actors are rationally pursuing some goals

Examples?

Optimization assumptions generate precise, solvable models

Optimization models appear to be performing fairly well in explaining reality

2 Economic Models: General Features

3. Positive and Normative Economics

- Establishing cause-and-effect relationships to create theories is known as positive economics. **Positive economics** is relatively scientific (testable) and focuses on value-free descriptions of and predictions about economic relationships.
- **Normative economics** deals with values and addresses what **ought to be** rather than what is. Sometimes economists refer to this as policy economics. Thus, they act to apply economic theories to policy. This is where economists usually differ from each other.

Once positive economics establishes the rules, normative economics seeks to apply them.

Public transport subsidy: Is this policy fair?

TABLA 7.1 | INGRESOS Y EGRESOS SISTEMA | 2014-2024¹

¹ En anexos, tablas y gráficos puede encontrar los ingresos y egresos del Sistema desde el año 2014.

Año	En Base Devengada		
	Ingresos Operacionales	Ingresos por Subsidio Ley 20.378	Egresos
2014	\$ 882.298	\$ 705.246	\$ 1.587.544
2015	\$ 901.950	\$ 579.147	\$ 1.481.097
2016	\$ 820.888	\$ 653.298	\$ 1.474.186
2017	\$ 819.104	\$ 728.321	\$ 1.547.425
2018	\$ 878.522	\$ 806.662	\$ 1.685.184
2019	\$ 838.172	\$ 893.555	\$ 1.731.727
2020	\$ 321.565	\$ 845.818	\$ 1.167.383
2021	\$ 417.277	\$ 780.961	\$ 1.198.238
2022	\$ 502.898	\$ 969.255	\$ 1.472.153
2023	\$ 510.955	\$ 1.077.214	\$ 1.588.169
2024	\$ 544.384	\$ 1.111.704	\$ 1.656.088

Nota: Valores expresado en millones y actualizados al valor presente de la UF del 31 de diciembre 2024.

Fuente: [Informe de Gestión 2024](#) (DTPM), valores en millones de pesos de dic 2024. 24

Partial Equilibrium Models versus General Equilibrium Models

Marshallian Supply-Demand Synthesis

- Alfred Marshall showed that supply and demand simultaneously operate to determine price
- Prices reflect both the marginal evaluation that consumers place on goods and the marginal costs of producing the goods
- Diamond-water paradox

Partial Equilibrium Models versus General Equilibrium Models

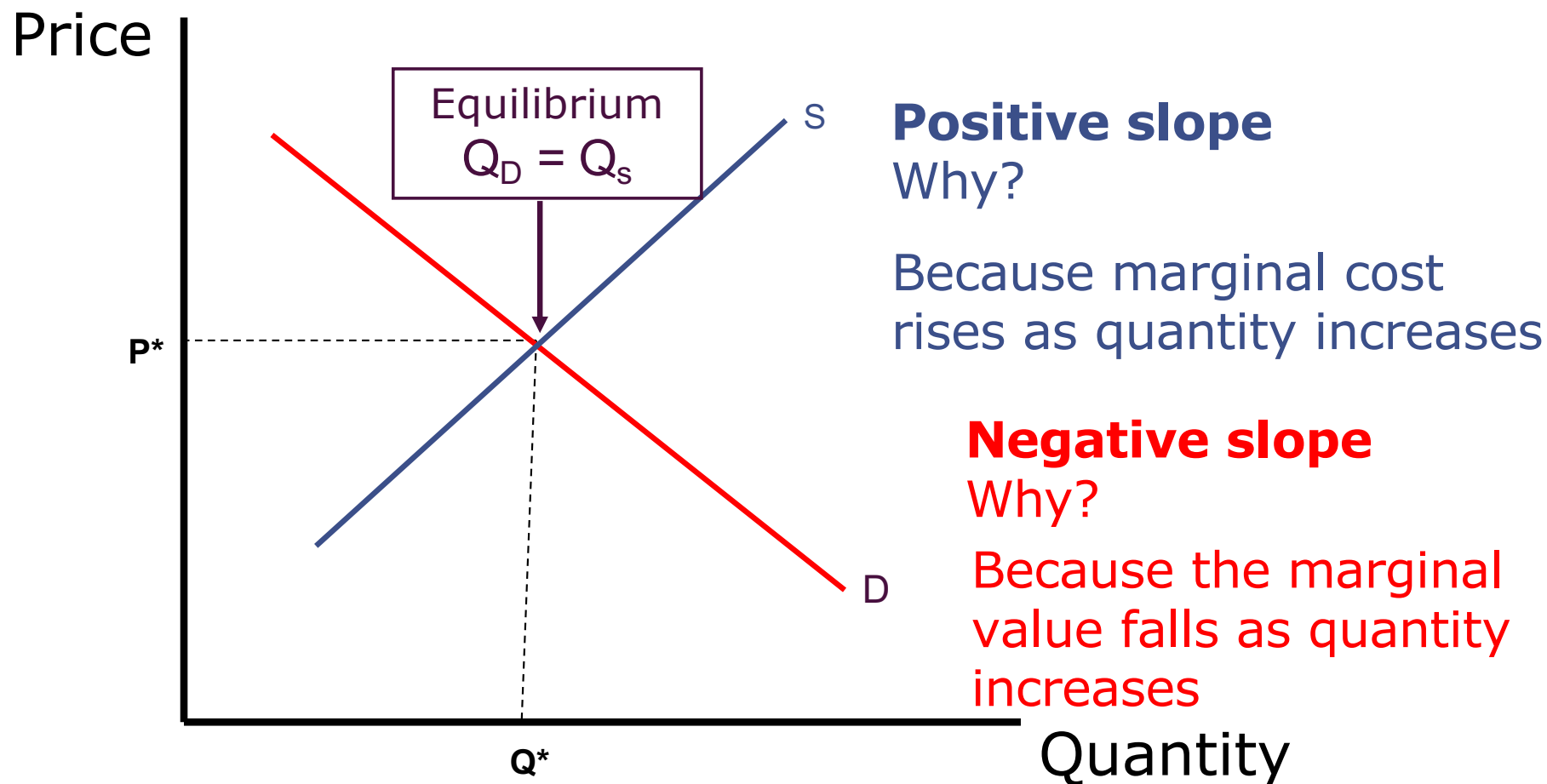
Water-Diamond Paradox

$$\frac{MgU_{Diamonds}}{P_{Diamonds}} = \frac{MgU_{Water}}{P_{Water}}$$

$$\frac{MgU_{Diamonds}}{MgU_{Water}} = \frac{P_{Diamonds}}{P_{Water}}$$

Partial Equilibrium Models versus General Equilibrium Models

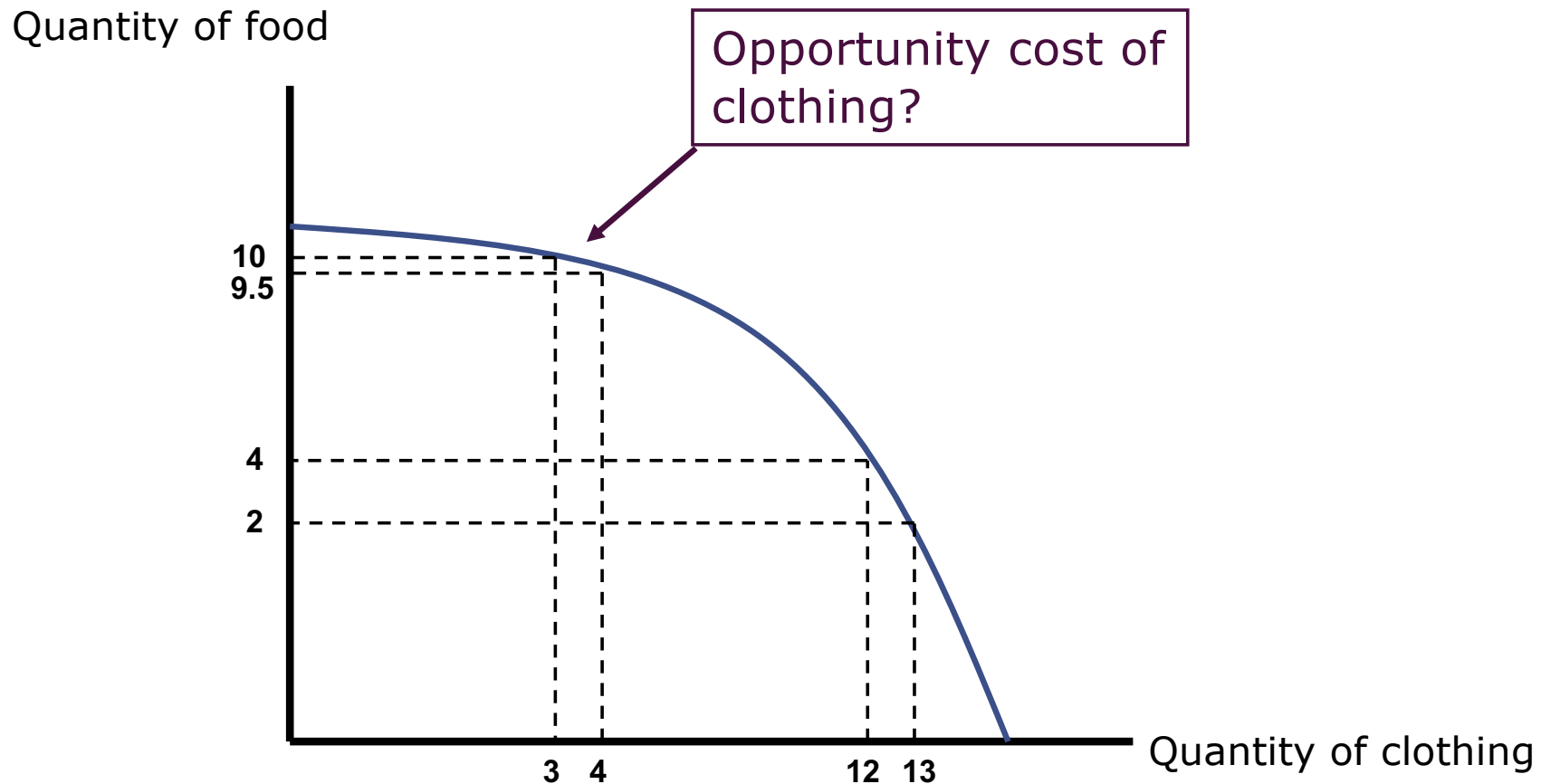
Supply-Demand Equilibrium



Partial Equilibrium Models versus General Equilibrium Models

- Marshallian Supply-Demand Equilibrium is a **partial equilibrium model**, since it focuses in only one market at a time
- For general questions – answers
 - We need a model for the entire economy
 - We need to include the interrelationships between markets and economic agents
- The **production possibilities frontier (PPF)** can be used as a basic building block for general equilibrium models
- A PPF shows the combinations of two outputs that can be produced with an economy's resources

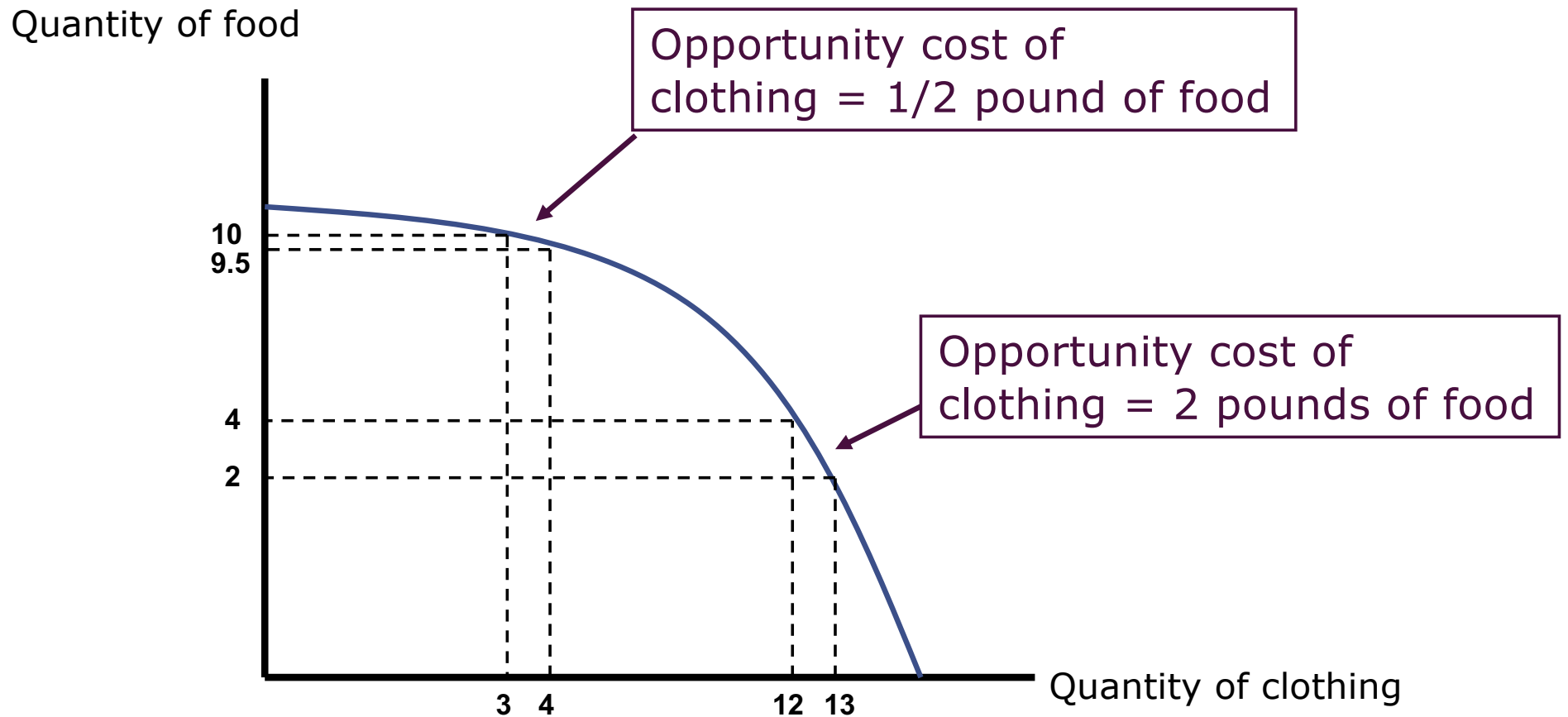
Production Possibilities Frontier (PPF)



PPF's slope rises as y rises

PPF is concave and its slope is negative

Production Possibilities Frontier (PPF)



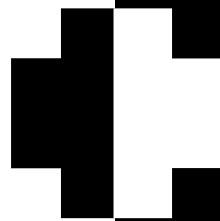
PPF 's slope rises as y rises

Partial Equilibrium Models versus General Equilibrium Models

PPF

- The production possibility frontier reminds us that **resources are scarce**
- Scarcity means that we must make choices
 - Each choice has opportunity costs
 - The opportunity costs depend on how much of each good is produced
 - Points in PPF are efficient

**Consumer
Theory**



**Producer
Theory**

Welfare Economics and Competitive Markets

Market Failures

Game Theory

One-minute paper - Canvas

Write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.